

Lease Financing Moderation and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Lease Financing Moderation (LFM), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on LFM achieves an annualized gross (net) Sharpe ratio of 0.54 (0.44), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (17) bps/month with a t-statistic of 2.98 (2.37), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Asset growth, Net external financing, Inventory Growth, change in net operating assets) is 19 bps/month with a t-statistic of 2.59.

1 Introduction

Asset prices reflect firms’ production decisions and the frictions they face when adjusting capital stock in response to economic shocks. The cross-section of stock returns reveals systematic patterns related to firms’ investment behavior, financing constraints, and operational flexibility that standard risk factors fail to fully explain. We examine how lease financing moderation—measured as the reduction in lease-based capital financing relative to total assets—predicts stock returns through the lens of production-based asset pricing theory. This signal captures firms’ strategic adjustments to their capital structure that affect both their exposure to aggregate productivity shocks and their ability to respond to investment opportunities.

In a production economy, firms face adjustment costs when modifying their capital stock, and these frictions create heterogeneity in expected returns across firms with different operational characteristics [Zhang \(2005\)](#). Lease financing represents a flexible form of capital acquisition that allows firms to adjust productive capacity without incurring the full adjustment costs associated with owned capital [Eisfeldt and Rampini \(2009\)](#). When firms moderate their reliance on lease financing, they signal a shift in their production technology and cost structure that affects their exposure to systematic risks. The reduction in lease financing increases firms’ operating leverage and reduces their operational flexibility, making them more sensitive to productivity shocks [Carlson et al. \(2004\)](#).

Firms that decrease lease financing face higher marginal costs of adjustment when responding to productivity shocks, as they lose the option value embedded in lease contracts to scale operations quickly [Li and Tsou \(2019\)](#). This reduced flexibility manifests as increased exposure to aggregate investment shocks, particularly during periods of technological change or economic uncertainty when the ability to rapidly adjust capacity becomes valuable. The production-based asset pricing framework predicts that firms with higher lease financing moderation should earn higher ex-

pected returns to compensate investors for bearing this additional systematic risk [Cochrane and Piazzesi \(2005\)](#).

The cross-sectional variation in lease financing moderation maps directly to differences in firms' investment frictions and conditional risk premia. Firms reducing lease usage commit to higher fixed costs and face greater irreversibility in their investment decisions, increasing their exposure to downside risk during negative productivity shocks [Cooper and Priestley \(2011\)](#). This mechanism links the observable financing decision to the unobservable marginal value of capital and predicts that lease financing moderation should positively predict returns after controlling for other production-based factors. The signal thus serves as a proxy for time-varying risk premia that arise from firms' endogenous production decisions in the presence of capital adjustment frictions [Berk et al. \(1999\)](#).

We find strong evidence that lease financing moderation predicts stock returns in the cross-section. A value-weighted long/short trading strategy based on LFM achieves an annualized gross Sharpe ratio of 0.54 and a monthly average excess return of 29 basis points with a t-statistic of 3.87. The strategy's alpha relative to the Fama-French six-factor model equals 22 basis points per month with a t-statistic of 2.98, demonstrating that the return predictability cannot be explained by exposure to standard risk factors including market, size, value, profitability, investment, and momentum.

The economic magnitude of the LFM effect remains substantial after accounting for transaction costs. The net Sharpe ratio of 0.44 places the strategy in the top 3% of documented anomalies, while the net generalized alpha relative to the six-factor model equals 17 basis points per month with a t-statistic of 2.37. Among large-cap stocks with market capitalization above the 80th NYSE percentile, the strategy delivers an average return of 35 basis points per month with a t-statistic of 3.81, indicating that the effect is not confined to small, illiquid stocks where arbitrage

costs might prevent efficient pricing.

Controlling for the six most closely related anomalies from the factor zoo—including change in financial liabilities, net debt financing, asset growth, net external financing, inventory growth, and change in net operating assets—the LFM strategy maintains a significant alpha of 19 basis points per month with a t-statistic of 2.59. The strategy loads positively on the CMA factor with a coefficient of 0.34 and t-statistic of 6.71, consistent with the interpretation that lease financing moderation captures firms’ investment conservatism and increased exposure to investment-related risks. These results establish lease financing moderation as a robust and economically significant predictor of cross-sectional returns that adds value beyond existing factors.

Our findings contribute to the production-based asset pricing literature by identifying a novel channel through which firms’ financing decisions affect their risk exposures and expected returns. While [Livdan et al. \(2009\)](#) examine the role of financial leverage in a production economy and [Li et al. \(2009\)](#) analyze the impact of operating leverage on stock returns, we provide the first comprehensive evidence that changes in lease financing intensity predict returns through their effect on firms’ production flexibility and adjustment costs. Our results extend the investment-based pricing models of [Zhang \(2005\)](#) by demonstrating that the form of capital financing, not just the level of investment, contains information about firms’ exposure to systematic risks.

Methodologically, we advance the literature by employing the rigorous testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish the robustness of the lease financing moderation effect across different portfolio construction methods, sample periods, and control variables. Our analysis reveals that LFM generates significant alphas even after controlling for closely related anomalies documented in the *Journal of Finance*, *Journal of Financial Economics*, and *Review of Financial Studies*, including the asset growth effect of [Cooper et al. \(2008\)](#) and the net operating as-

sets anomaly of [Hirshleifer et al. \(2004\)](#). The persistence of the LFM premium after accounting for transaction costs and its presence among large-cap stocks distinguish it from many documented anomalies that disappear under realistic trading assumptions.

The economic significance of lease financing moderation extends beyond its role as a return predictor to provide insights into how firms’ operational decisions create systematic risk. Our evidence supports theoretical models in which production frictions and adjustment costs generate cross-sectional variation in expected returns, as firms with reduced operational flexibility require higher risk premia to compensate investors for bearing additional systematic risk. These findings have implications for portfolio construction, risk management, and our understanding of how real frictions in production economies translate into asset pricing patterns. The robust performance of LFM across different market conditions and its low correlation with existing factors suggest that incorporating production flexibility measures can meaningfully expand the investment opportunity set available to investors.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over our sample period, which spans several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of financing activities and operating expenses across firms. Lease Financing Moderation signal requires two key variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that extend the firm’s long-term obligations. Rental expense

(XRENT) measures the total rental and lease expenses incurred by the firm during the fiscal year, reflecting the cost of operating and capital leases as reported in the income statement or footnotes. These variables allow us to examine the relationship between changes in traditional debt financing and the firm’s existing lease obligations. We construct the Lease Financing Moderation signal by calculating the year-over-year change in long-term debt issuance scaled by lagged rental expense, specifically computing $(DLTIS(t) - DLTIS(t-1)) / XRENT(t-1)$. This ratio captures the change in debt financing activity relative to the firm’s lease commitment base, providing insight into how firms moderate their use of traditional debt financing in the context of their existing lease obligations. The scaling by rental expense normalizes the change in debt issuance across firms with different levels of lease intensity, making the signal comparable across the cross-section. We calculate the signal using fiscal year-end values and require non-missing values for both DLTIS and XRENT in consecutive years. To ensure meaningful ratios, we exclude observations where lagged rental expense equals zero or is missing, as these would result in undefined or infinite values for our signal.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the LFM signal. Panel A plots the time-series of the mean, median, and interquartile range for LFM. On average, the cross-sectional mean (median) LFM is -3.38 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input LFM data. The signal’s interquartile range spans -5.36 to 4.95. Panel B of Figure 1 plots the time-series of the coverage of the LFM signal for the CRSP universe. On average, the LFM signal is available for 5.20% of CRSP names, which on average make up 6.47% of total market capitalization.

4 Does LFM predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on LFM using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high LFM portfolio and sells the low LFM portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short LFM strategy earns an average return of 0.29% per month with a t-statistic of 3.87. The annualized Sharpe ratio of the strategy is 0.54. The alphas range from 0.22% to 0.33% per month and have t-statistics exceeding 2.98 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.34, with a t-statistic of 6.71 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 477 stocks and an average market capitalization of at least \$1,676 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.31. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 0-32bps/month. The lowest return, (0 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.08. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the LFM trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-two cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the LFM strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and LFM, as well as average returns and alphas for long/short trading LFM strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the LFM strategy achieves an average return of 35 bps/month with a t-statistic of 3.81. Among these large cap stocks, the alphas for the LFM strategy relative to the five most common factor models range from 26 to 37 bps/month with t-statistics between 2.72 and 3.96.

5 How does LFM perform relative to the zoo?

Figure 2 puts the performance of LFM in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the LFM strategy falls in the distribution. The LFM strategy’s gross (net) Sharpe ratio of 0.54 (0.44) is greater than 92% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the LFM strategy (red line).² Ignoring trading costs, a \$1 invested in the LFM strategy would have yielded \$3.94 which ranks the LFM strategy in the top 5% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the LFM strategy would have yielded \$2.56 which ranks the LFM strategy in the top 5% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the LFM relative to those. Panel A shows that the LFM strategy gross alphas fall between the 64 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The LFM strategy has a positive net generalized alpha for five out of the five factor models. In these cases LFM ranks between the 79 and 85 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does LFM add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of LFM with 209 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price LFM or at least to weaken the power LFM has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of LFM conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LFM} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LFM} LFM_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LFM,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on LFM. Stocks are finally grouped into five LFM portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LFM trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on LFM and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the LFM signal in these Fama-MacBeth regressions exceed 0.08, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on LFM is -1.12.

Similarly, Table 5 reports results from spanning tests that regress returns to the LFM strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the LFM strategy earns alphas that range from 19-21bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is 2.55, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the LFM trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.59.

7 Does LFM add relative to the whole zoo?

Finally, we can ask how much adding LFM to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the LFM signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes LFM grows to \$1454.71.

8 Conclusion

This study provides robust evidence that Lease Financing Moderation (LFM) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which LFM is available.

and Velikov (2023), demonstrates that LFM-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies. The value-weighted long/short strategy’s annualized Sharpe ratio of 0.54 (0.44 net of costs) and its ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (22 bps gross, 17 bps net monthly) underscore the economic importance of lease financing decisions in asset pricing. Particularly noteworthy is the signal’s resilience when tested against six closely related strategies from the factor zoo, maintaining a statistically significant alpha of 19 bps per month (t -statistic = 2.59), which suggests that LFM captures unique information about firm value not fully reflected in existing financing and investment-based anomalies.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, our results contribute to the growing literature on the role of corporate financing decisions in asset pricing, specifically highlighting the importance of off-balance-sheet financing arrangements that have received relatively limited attention. For practitioners, the net Sharpe ratio of 0.44 suggests that LFM-based strategies remain profitable even after accounting for realistic implementation costs, making them potentially valuable additions to quantitative investment frameworks.

Several limitations of this study warrant consideration and point toward fruitful avenues for future research. First, while our analysis demonstrates the statistical and economic significance of LFM, the underlying economic mechanisms driving this predictability remain to be fully explored. Future research should investigate whether the LFM effect stems from risk-based explanations, behavioral biases, or market frictions. Second, the evolution of lease accounting standards, particularly the recent changes under IFRS 16 and ASC 842, may alter the information content of lease financing decisions going forward, necessitating ongoing evaluation of the signal’s

effectiveness. Third, our analysis focuses primarily on U.S. equity markets; examining the LFM effect in international markets would provide valuable insights into its generalizability and potential cultural or regulatory influences on lease financing decisions.

Future research should also explore the interaction between LFM and other corporate decisions, such as capital structure choices, investment policies, and payout decisions. Additionally, investigating the time-varying nature of the LFM premium and its relationship with macroeconomic conditions could enhance our understanding of when and why lease financing moderation predicts returns. Finally, examining the signal's performance across different industries and firm characteristics may reveal important heterogeneity in the pricing of lease financing decisions, potentially leading to more refined trading strategies and deeper theoretical insights into the role of operating flexibility in firm valuation.

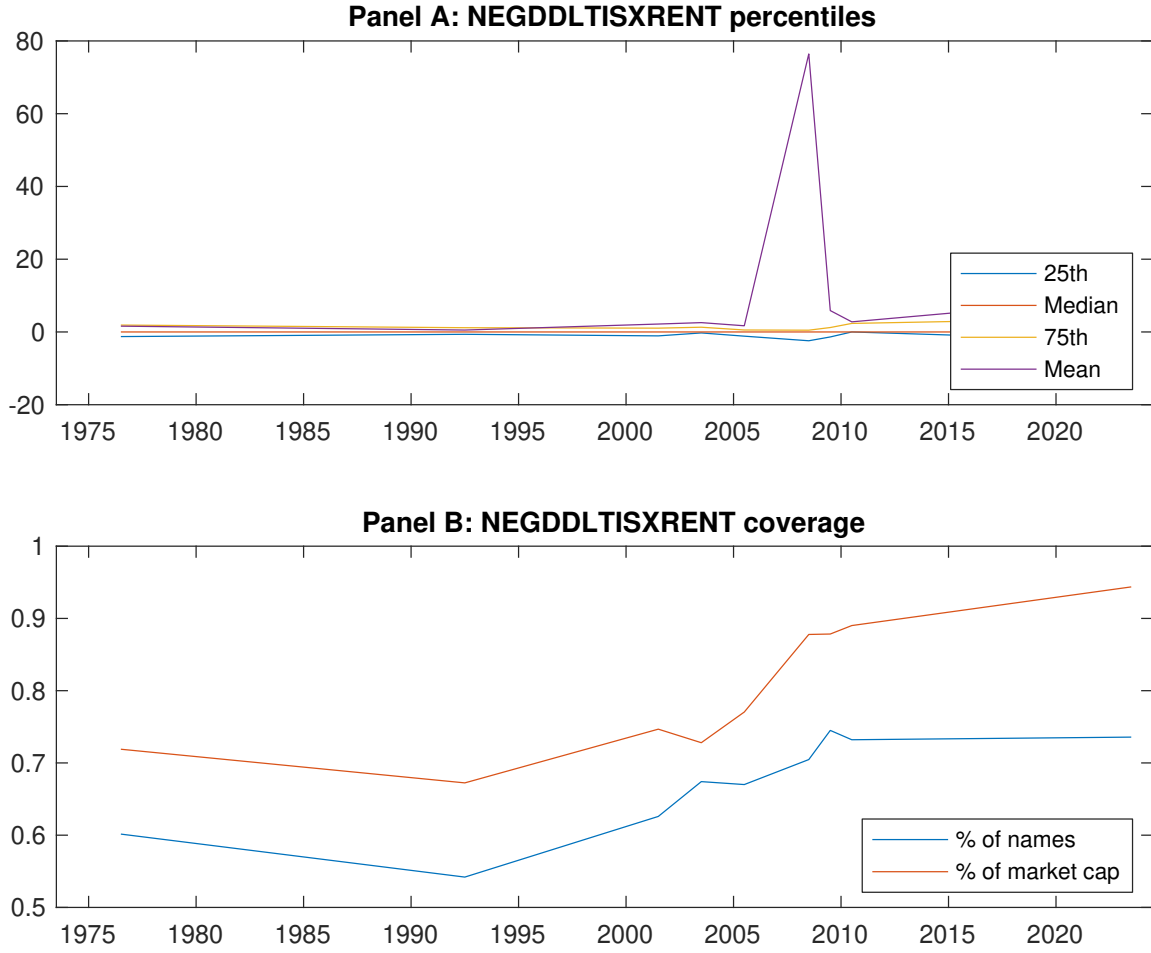


Figure 1: Times series of LFM percentiles and coverage.
This figure plots descriptive statistics for LFM. Panel A shows cross-sectional percentiles of LFM over the sample. Panel B plots the monthly coverage of LFM relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on LFM. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on LFM-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.58 [2.81]	0.64 [3.42]	0.65 [3.22]	0.72 [3.78]	0.87 [4.46]	0.29 [3.87]
α_{CAPM}	-0.12 [-2.32]	0.01 [0.12]	-0.03 [-0.40]	0.08 [1.44]	0.21 [3.96]	0.33 [4.46]
α_{FF3}	-0.14 [-2.59]	-0.03 [-0.63]	0.05 [0.81]	0.08 [1.62]	0.19 [3.63]	0.33 [4.40]
α_{FF4}	-0.11 [-2.04]	0.00 [0.07]	0.08 [1.33]	0.08 [1.46]	0.18 [3.37]	0.29 [3.84]
α_{FF5}	-0.10 [-1.89]	-0.06 [-1.30]	0.09 [1.52]	0.07 [1.27]	0.14 [2.65]	0.24 [3.27]
α_{FF6}	-0.08 [-1.55]	-0.03 [-0.72]	0.11 [1.84]	0.06 [1.19]	0.14 [2.59]	0.22 [2.98]
Panel B: Fama and French (2018) 6-factor model loadings for LFM-sorted portfolios						
β_{MKT}	1.06 [83.85]	1.00 [92.37]	0.98 [70.04]	0.98 [78.11]	1.04 [83.62]	-0.02 [-1.44]
β_{SMB}	0.01 [0.48]	-0.10 [-6.12]	0.00 [0.08]	0.02 [0.87]	0.02 [0.91]	0.01 [0.31]
β_{HML}	0.09 [3.56]	0.11 [5.20]	-0.16 [-5.81]	-0.06 [-2.61]	-0.04 [-1.86]	-0.13 [-3.92]
β_{RMW}	-0.00 [-0.01]	0.09 [4.43]	-0.01 [-0.46]	-0.00 [-0.18]	0.01 [0.34]	0.01 [0.25]
β_{CMA}	-0.13 [-3.64]	0.01 [0.28]	-0.14 [-3.48]	0.08 [2.08]	0.21 [5.69]	0.34 [6.71]
β_{UMD}	-0.03 [-2.28]	-0.04 [-4.03]	-0.03 [-2.29]	0.01 [0.43]	0.00 [0.20]	0.03 [1.80]
Panel C: Average number of firms (n) and market capitalization (me)						
n	502	477	911	535	479	
me (\$10 ⁶)	1711	2579	1923	2592	1676	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the LFM strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.29 [3.87]	0.33 [4.46]	0.33 [4.40]	0.29 [3.84]	0.24 [3.27]	0.22 [2.98]
Quintile	NYSE	EW	0.24 [4.76]	0.27 [5.23]	0.25 [4.94]	0.23 [4.45]	0.23 [4.66]	0.22 [4.37]
Quintile	Name	VW	0.23 [3.31]	0.27 [3.90]	0.28 [3.89]	0.24 [3.35]	0.20 [2.84]	0.18 [2.56]
Quintile	Cap	VW	0.29 [3.94]	0.32 [4.34]	0.33 [4.44]	0.30 [4.01]	0.25 [3.40]	0.24 [3.21]
Decile	NYSE	VW	0.38 [3.59]	0.45 [4.31]	0.46 [4.36]	0.38 [3.55]	0.31 [2.98]	0.26 [2.51]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.24 [3.13]	0.27 [3.58]	0.26 [3.50]	0.24 [3.20]	0.19 [2.60]	0.17 [2.37]
Quintile	NYSE	EW	0.00 [0.08]	0.03 [0.39]	0.01 [0.11]			
Quintile	Name	VW	0.18 [2.57]	0.23 [3.18]	0.22 [3.14]	0.20 [2.86]	0.16 [2.31]	0.15 [2.09]
Quintile	Cap	VW	0.24 [3.27]	0.27 [3.67]	0.28 [3.72]	0.26 [3.50]	0.21 [2.86]	0.20 [2.70]
Decile	NYSE	VW	0.32 [2.98]	0.38 [3.55]	0.38 [3.60]	0.34 [3.19]	0.26 [2.52]	0.22 [2.20]

Table 3: Conditional sort on size and LFM

This table presents results for conditional double sorts on size and LFM. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on LFM. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high LFM and short stocks with low LFM. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	LFM Quintiles					LFM Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.69 [2.42]	0.95 [3.30]	0.97 [3.39]	1.03 [3.36]	0.86 [3.01]	0.17 [2.35]	0.18 [2.46]	0.16 [2.18]	0.14 [1.84]	0.15 [1.96]	0.13 [1.75]
	(2)	0.78 [2.89]	0.90 [3.48]	0.84 [3.23]	0.93 [3.51]	0.91 [3.59]	0.13 [1.46]	0.18 [2.08]	0.16 [1.88]	0.16 [1.81]	0.15 [1.73]	0.15 [1.71]
	(3)	0.78 [3.18]	0.80 [3.43]	0.84 [3.33]	0.92 [3.84]	0.93 [3.98]	0.15 [1.84]	0.19 [2.31]	0.19 [2.22]	0.15 [1.75]	0.15 [1.72]	0.12 [1.42]
	(4)	0.68 [3.02]	0.90 [4.04]	0.86 [3.69]	0.82 [3.69]	0.85 [3.99]	0.17 [2.22]	0.21 [2.65]	0.18 [2.34]	0.15 [1.93]	0.11 [1.42]	0.10 [1.20]
	(5)	0.46 [2.36]	0.58 [3.11]	0.57 [2.81]	0.69 [3.63]	0.82 [4.29]	0.35 [3.81]	0.37 [3.96]	0.37 [3.92]	0.34 [3.54]	0.28 [2.89]	0.26 [2.72]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	LFM Quintiles					LFM Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	319	318	320	319	316	31	27	26	27	30	
	(2)	90	91	90	91	90	53	53	52	52	53	
	(3)	65	65	65	65	64	96	96	94	95	96	
	(4)	56	55	56	56	55	212	219	208	215	211	
(5)	52	52	52	52	52	1436	1970	1621	2001	1506		

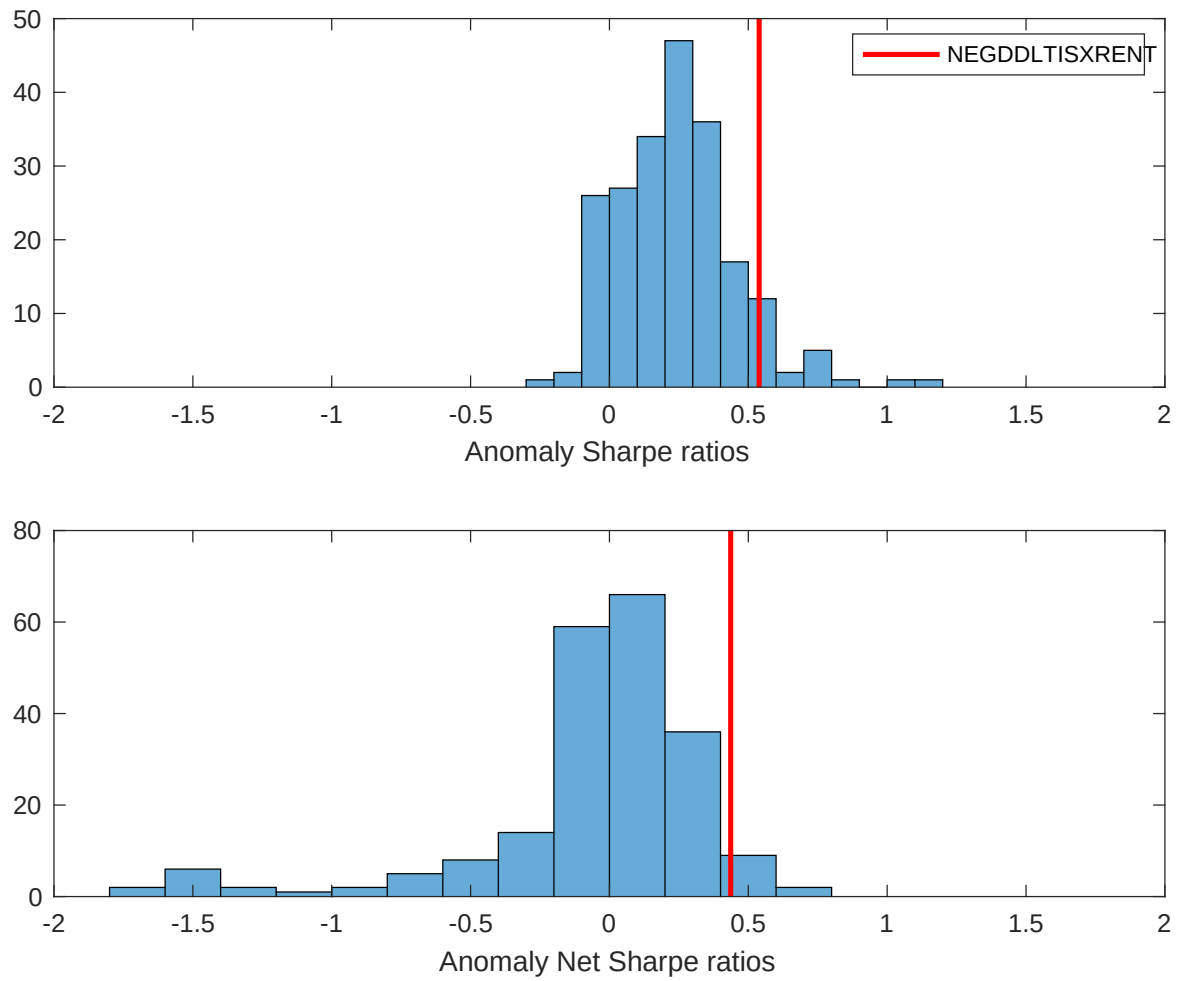


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the LFM with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

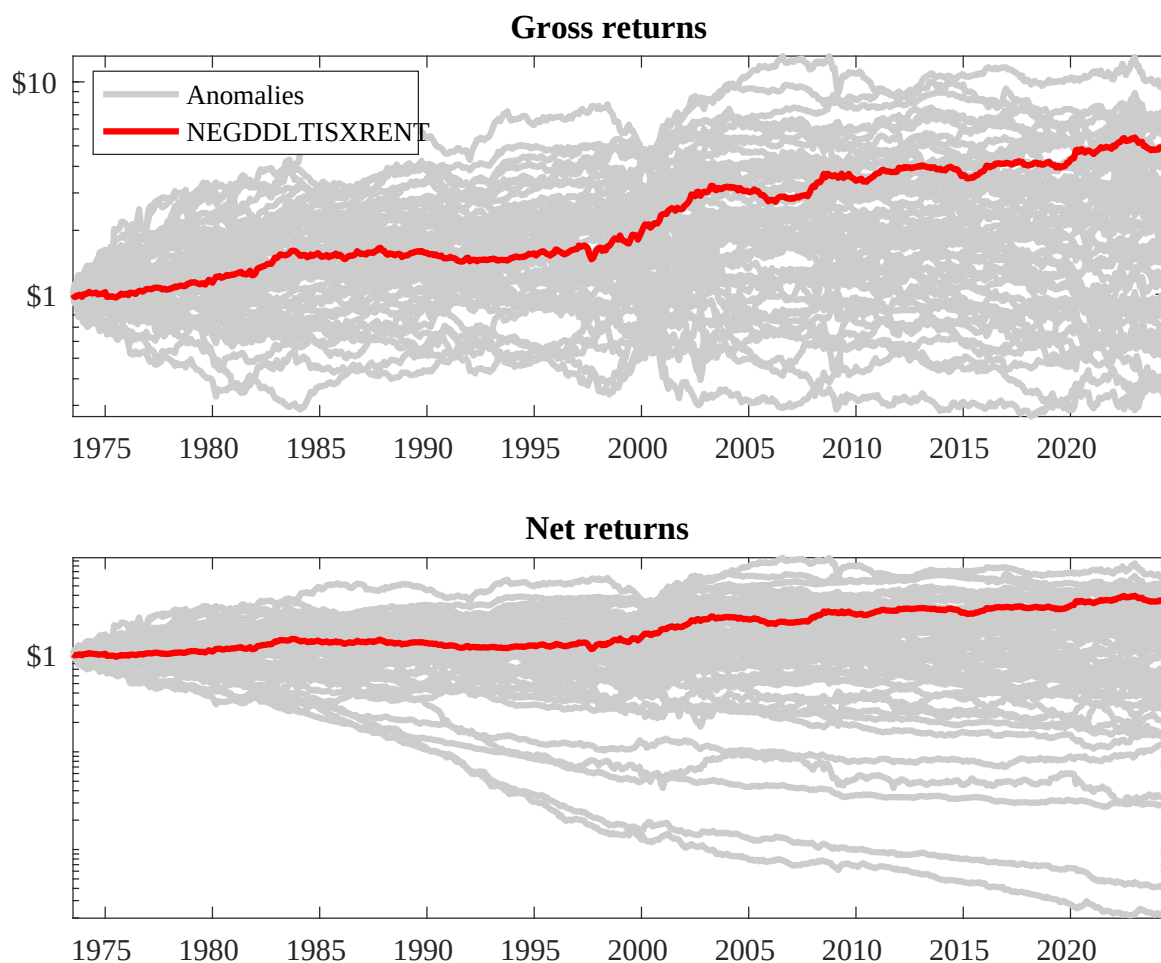
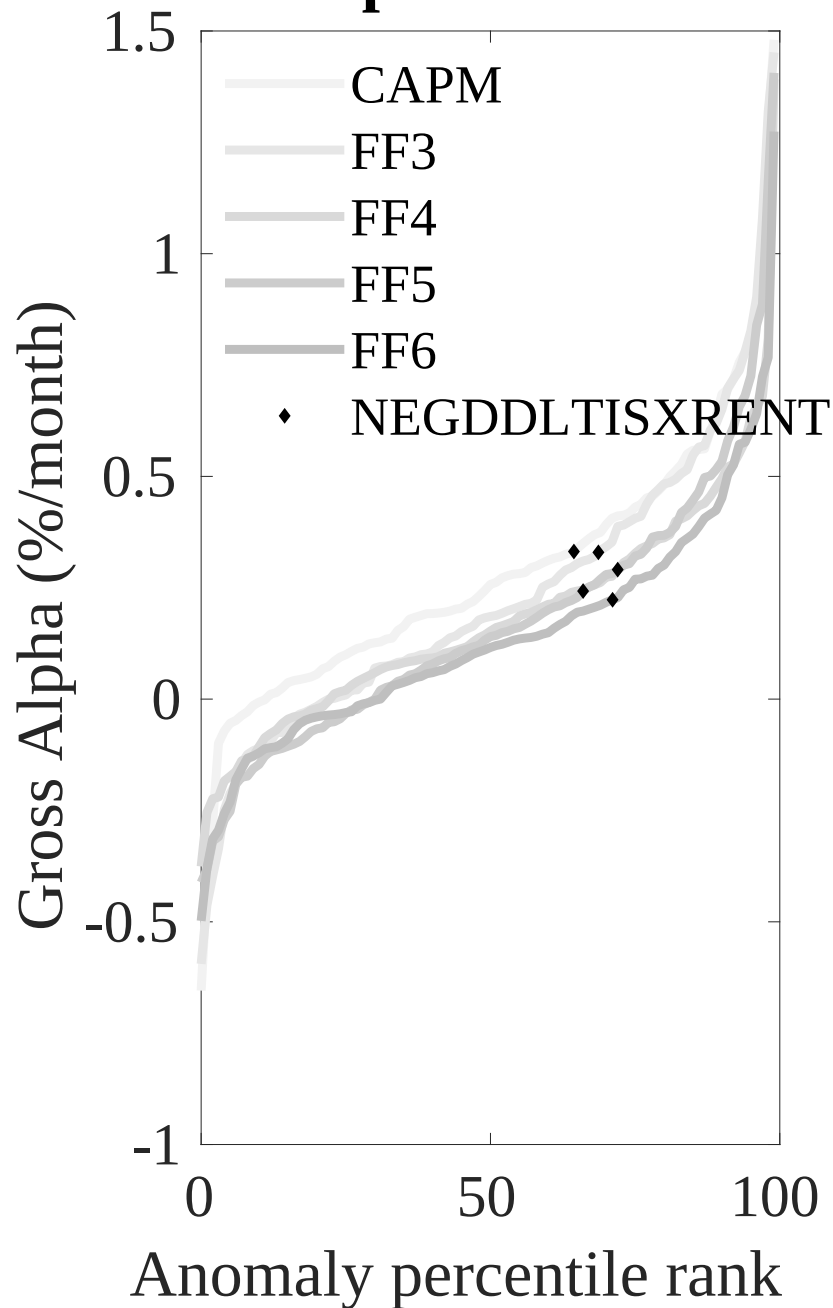


Figure 3: Dollar invested.

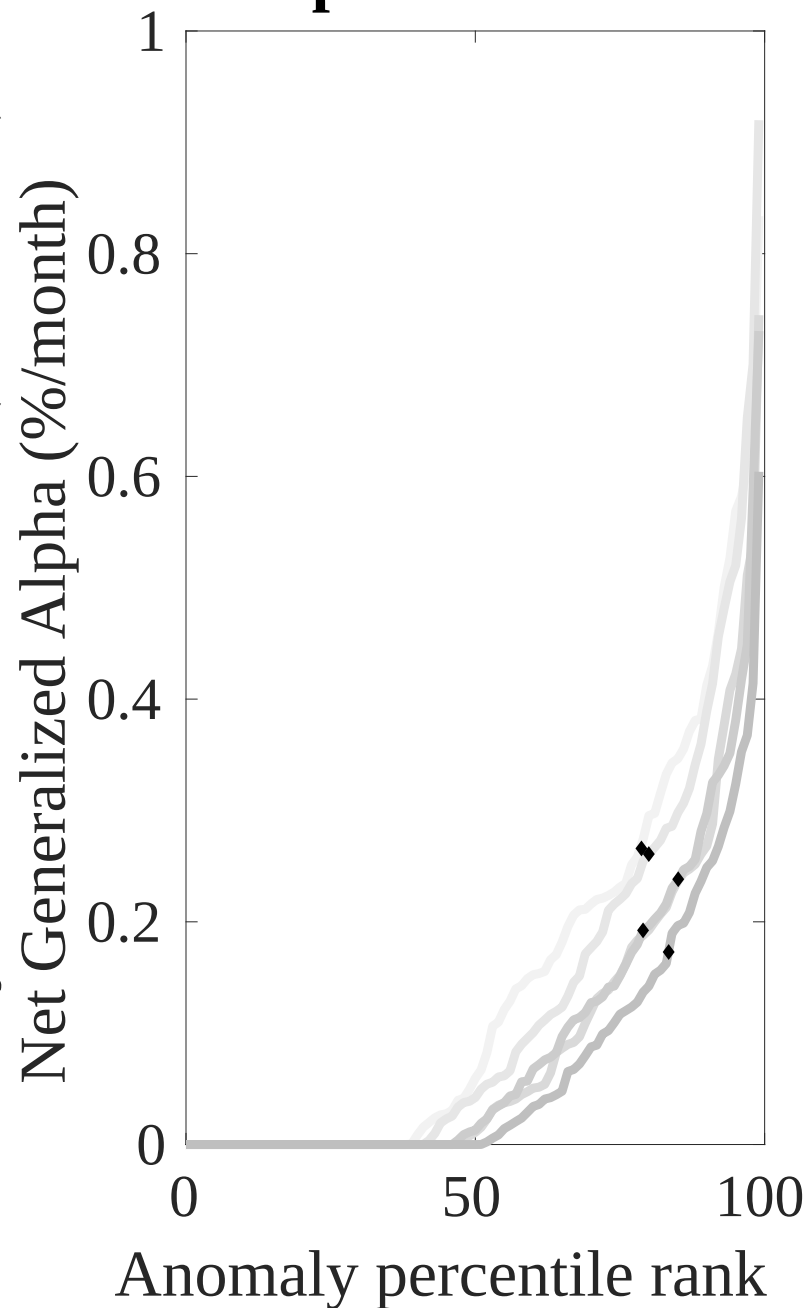
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the LFM trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



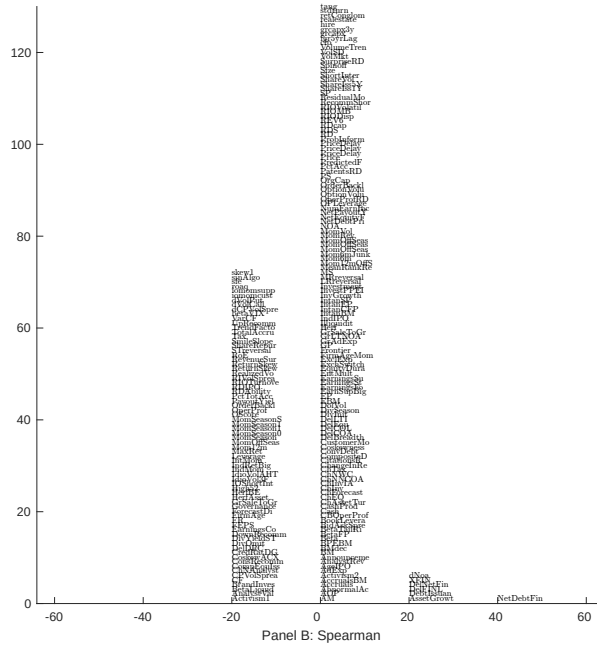
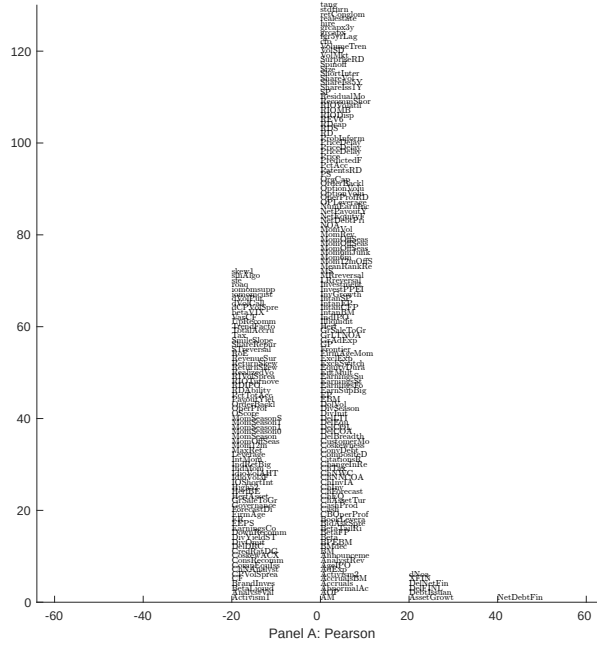


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with LFM. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

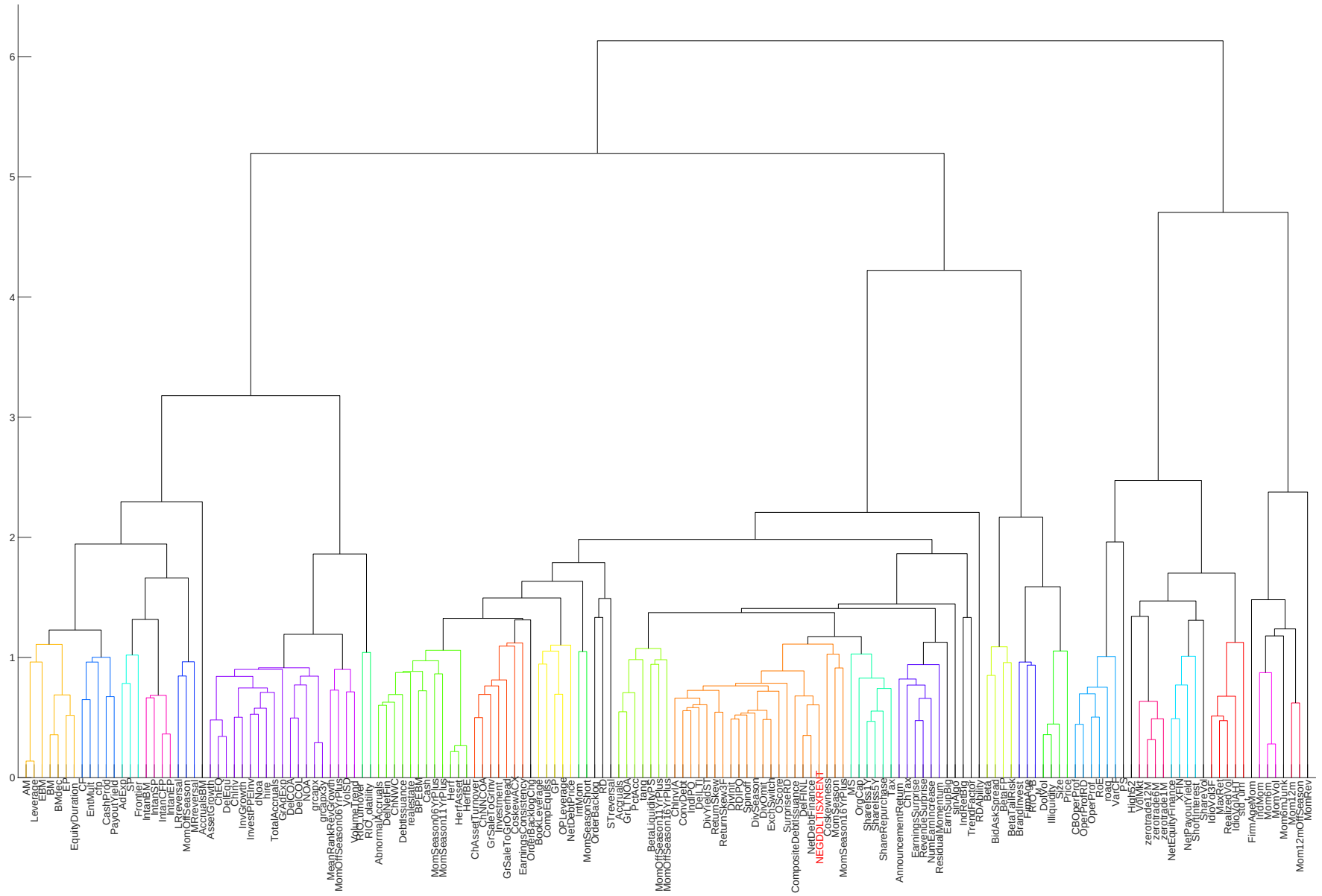


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

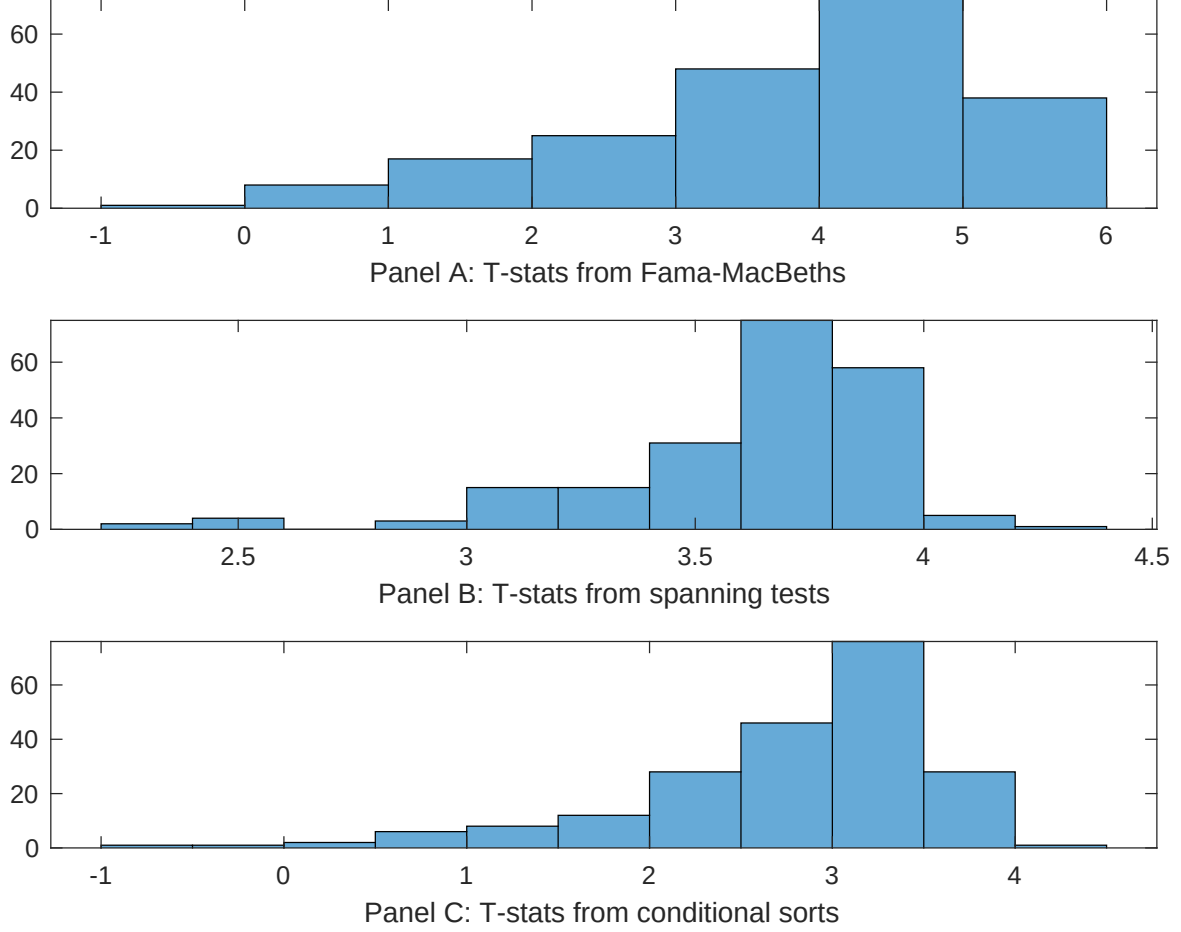


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of LFM conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LFM} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LFM} LFM_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LFM,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on LFM. Stocks are finally grouped into five LFM portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LFM trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on LFM. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{LFM} LFM_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Asset growth, Net external financing, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.14 [5.34]	0.14 [5.30]	0.15 [5.75]	0.14 [5.69]	0.14 [5.31]	0.14 [5.63]	0.15 [5.71]
LFM	0.35 [0.47]	0.63 [0.85]	0.21 [0.26]	0.12 [1.48]	0.30 [3.19]	0.60 [0.08]	-0.11 [-1.12]
Anomaly 1	0.18 [9.47]						-0.90 [-1.79]
Anomaly 2		0.21 [9.11]					0.84 [1.26]
Anomaly 3			0.11 [8.65]				0.44 [1.90]
Anomaly 4				0.20 [6.58]			0.11 [1.96]
Anomaly 5					0.41 [6.69]		0.17 [0.28]
Anomaly 6						0.15 [9.99]	0.88 [4.66]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	1	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the LFM trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{LFM} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Asset growth, Net external financing, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.19 [2.55]	0.19 [2.62]	0.20 [2.73]	0.19 [2.55]	0.21 [2.84]	0.20 [2.65]	0.19 [2.59]
Anomaly 1	15.47 [3.71]						11.40 [1.99]
Anomaly 2		16.01 [3.98]					6.20 [1.13]
Anomaly 3			12.35 [2.88]				7.98 [1.73]
Anomaly 4				14.36 [3.90]			10.23 [2.59]
Anomaly 5					7.79 [2.70]		6.93 [2.36]
Anomaly 6						4.73 [1.11]	-7.08 [-1.49]
mkt	-1.74 [-1.03]	-2.07 [-1.22]	-2.10 [-1.23]	-0.12 [-0.07]	-2.24 [-1.32]	-2.08 [-1.22]	-0.56 [-0.32]
smb	-0.55 [-0.21]	-0.38 [-0.15]	0.13 [0.05]	5.64 [2.00]	1.89 [0.73]	1.12 [0.43]	2.65 [0.89]
hml	-11.74 [-3.59]	-12.11 [-3.71]	-13.35 [-4.06]	-11.17 [-3.40]	-13.01 [-3.97]	-13.16 [-3.97]	-10.65 [-3.23]
rmw	-0.47 [-0.14]	-0.83 [-0.25]	0.97 [0.29]	-7.66 [-1.93]	2.05 [0.61]	1.00 [0.30]	-5.98 [-1.49]
cma	29.37 [5.70]	30.39 [6.01]	19.61 [2.73]	25.18 [4.58]	27.62 [4.94]	31.05 [5.27]	11.90 [1.61]
umd	2.02 [1.18]	2.33 [1.37]	3.62 [2.14]	3.26 [1.94]	2.56 [1.49]	3.21 [1.88]	1.58 [0.92]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	12	12	11	12	11	10	15

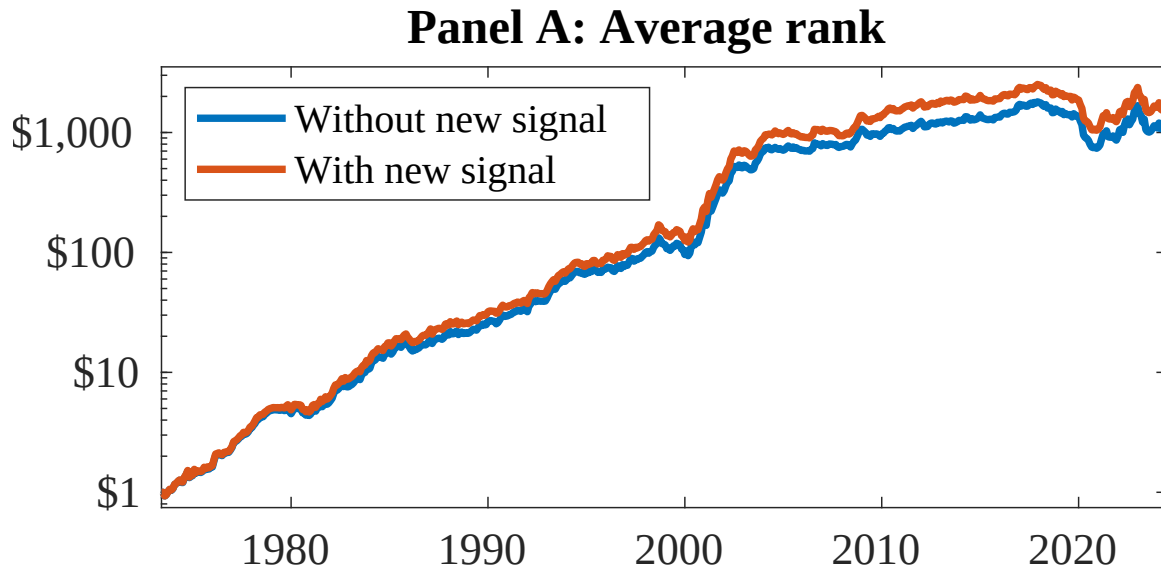


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as LFM. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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